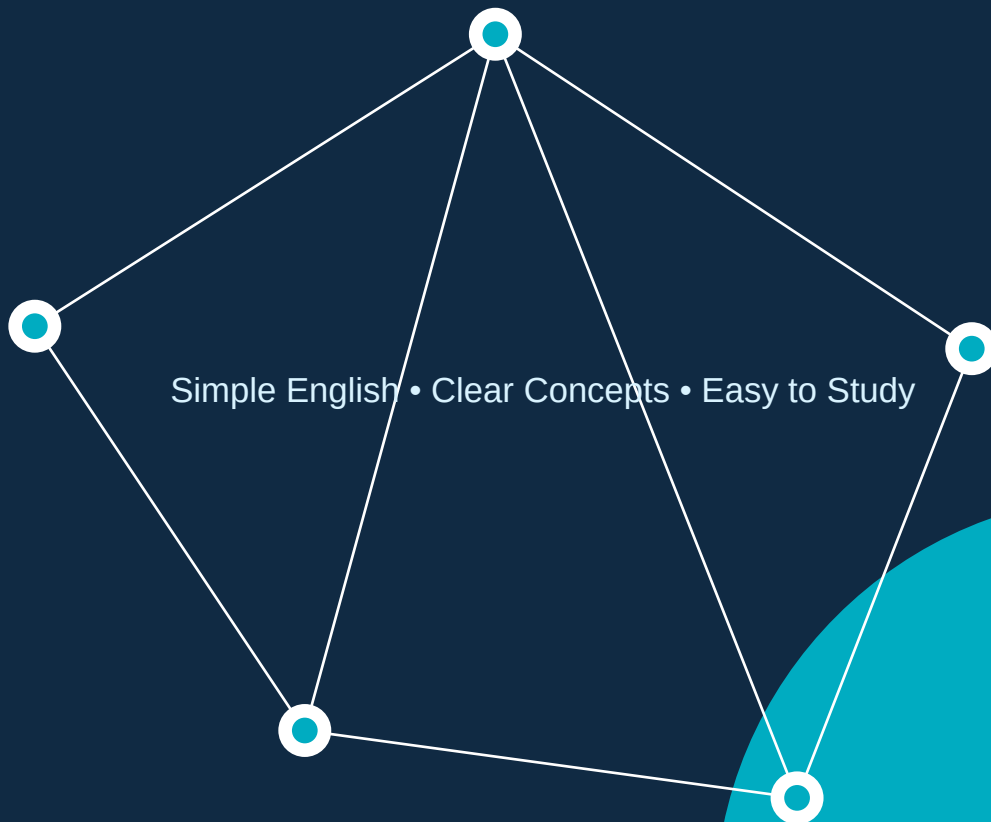


WEB DEVELOPMENT

A Complete Beginner's Guide

What it is • Why websites are used • How the Web works • History • HTML, CSS & JavaScript • Types • Key terms



1. What Is Web Development?

Web development is the process of creating, building, testing, and maintaining websites and web applications that people access through a web browser such as Chrome, Edge, Firefox, or Safari.

A website can be very simple, such as a page that displays information, or very complex, such as an online shopping system, banking portal, social media platform, or learning website.

The three basic technologies

Technology	Main Job	Easy Meaning
HTML	Structure	Creates headings, paragraphs, images, forms, buttons, tables, and other content.
CSS	Presentation	Controls colors, fonts, spacing, layout, borders, and appearance.
JavaScript	Behavior	Makes pages interactive and able to respond to user actions.

Web development vs. web design

Web design mainly focuses on how a website looks and how users experience it. **Web development** focuses on building the working website and its features. In real projects, these areas often work together.

2. Why Do We Use Websites?

Information: Websites provide news, articles, documentation, educational material, and company information.

Communication: Websites support contact forms, messaging systems, forums, and portals.

Business: Companies use websites to display products, accept orders, provide support, and reach customers.

Education: Schools and learning platforms use websites for lessons, assignments, tests, and study resources.

Entertainment: Websites can provide videos, music, games, stories, and other digital content.

Online Services: Users can book, register, apply, pay, or check information online.

Important idea

A website is not only a page with text and pictures. Modern websites can behave like applications. For example, a shopping website can let a user search for a product, add it to a cart, make a payment, and receive an order confirmation.

3. How Does the Web Work?

When you type a website address into a browser and press Enter, several steps happen very quickly. This simplified flow is enough for a beginner to understand the basic idea.

Step	Part	What happens?
1	User	You enter a website address or click a link.
2	Browser	The browser prepares a request for the required web resource.
3	DNS	The domain name is translated into an IP address.
4	Server	The browser sends a request to the web server.
5	Response	The server sends files or data back to the browser.
6	Browser	The browser reads HTML, CSS, and JavaScript and builds the page.

A simple example

Suppose you visit **example.com**. Your browser needs to find the server for that domain, request the page, receive the required resources, and display them. The browser then uses HTML to understand the structure, CSS to style it, and JavaScript to provide interactive behavior.

What is a browser?

A **web browser** is software used to access websites. Chrome, Firefox, Edge, and Safari are examples. The browser downloads web resources and converts them into the page you see on your screen.

What is a web server?

A **web server** is a computer or software system that receives requests and sends web resources or data back to clients. A server can store website files and can also run programs that process user requests.

What is DNS?

DNS means Domain Name System. Humans prefer names such as example.com, while networks use IP addresses. DNS helps translate a domain name into the IP address needed to reach the appropriate server.

4. HTTP and HTTPS

HTTP means Hypertext Transfer Protocol. It is a protocol used for communication between a web browser and a web server. **HTTPS** is HTTP protected with encryption using TLS. You will commonly see HTTPS in the address bar of modern websites.

The beginner idea is: **the browser requests resources, and the server responds with resources or data.**

5. Who Introduced the World Wide Web?

The **World Wide Web (WWW)** was invented by **Tim Berners-Lee**, a British computer scientist, while working at CERN. He proposed the Web in **1989** as a way for researchers to share information over a network.

The first website went online around **1991**. Berners-Lee also developed early technologies that made the Web work together: **HTML** for documents, **HTTP** for communication, and the basic idea of **URLs** for addressing resources.

The Web should not be confused with the Internet. The **Internet** is the global network infrastructure that connects computers and networks. The **Web** is a system of interlinked resources and applications that runs over the Internet using technologies such as HTTP.

Year	Milestone
1989	Tim Berners-Lee proposed the World Wide Web at CERN.
1990	Early Web software and the first Web server/browser work were developed.
1991	The Web became available beyond the initial development environment; the first website was made accessible.
1993	CERN made the Web software available on a royalty-free basis, helping wider adoption.

Note: The Web has several historical milestones and contributors. The timeline above is simplified for beginner study.

6. Front-End and Back-End

Front-end is the part of a website that runs in the browser and that the user can see and interact with. HTML, CSS, and JavaScript are the core beginner technologies for front-end development.

Back-end is the server-side part of an application. It can handle business logic, authentication, data processing, APIs, and communication with databases. Common back-end technologies include Python, Java, JavaScript/Node.js, PHP, C#, Ruby, and others.

Database stores organized information such as user accounts, products, orders, marks, messages, or posts. Examples include MySQL, PostgreSQL, MongoDB, and SQLite.

7. HTML, CSS and JavaScript — The Basic Foundation

A useful way to remember the three technologies is to compare a web page to a house:

HTML	CSS	JavaScript
Structure of the house	Paint, design and arrangement	Actions and behavior

HTML example

```
<h1>My Website</h1>
<p>Welcome to my website.</p>
```

HTML tells the browser what content exists and how it is structured.

CSS example

```
h1 { color: blue; font-size: 32px; }
```

CSS tells the browser how the content should look.

JavaScript example

```
button.onclick = function() { alert('Hello!'); };
```

JavaScript can respond to actions such as clicks, typing, scrolling, and form submission.

8. Types of Websites

Static Website: Content is mainly fixed and usually changes only when the developer edits the files.

Dynamic Website: Content can change based on users, databases, time, or other information.

E-commerce Website: Used to display products or services and support online purchasing.

Blog / News Website: Used to publish articles, posts, news, and regularly updated content.

Educational Website: Provides lessons, courses, quizzes, resources, or student services.

Web Application: A website that provides application-like functions, such as email, banking, project management, or booking.

9. Important Web Terms

Term	Meaning
Website	A collection of related web pages and resources available through a web address.
Web Page	A single document or resource displayed by a browser.
Web Browser	Software such as Chrome, Firefox, Edge, or Safari used to access the Web.
Domain Name	A human-friendly name used to reach a website, such as example.com.
URL	Uniform Resource Locator; an address that identifies a resource on the Web.
IP Address	A numerical network address used to identify a device or network endpoint.
Web Hosting	A service that provides server resources where website files or applications can be stored and served.
API	Application Programming Interface; a defined way for software systems to communicate.
Database	A system used to store and manage structured information.
Responsive Design	Design that adapts a website layout to different screen sizes such as phones, tablets, and computers.

10. How a Simple Website Is Built

1. Decide the purpose of the website. Example: a student profile page.
2. Create the HTML structure: headings, paragraphs, images, links, and forms.
3. Add CSS: colors, fonts, spacing, borders, layout, and responsive rules.
4. Add JavaScript if the page needs interaction, such as buttons or form validation.
5. Test the website in a browser.
6. Fix errors and improve the design.
7. Publish the website using a hosting service and connect a domain name if needed.
8. Maintain the website by updating content, fixing problems, and improving security.

Example project structure

my-website/



■■■ index.html

■■■ style.css

■■■ script.js

index.html = page structure

style.css = page design

script.js = page behavior

11. What Happens After a User Clicks a Button?

Consider a login button. A simplified flow is:

1. The user enters a username and password.
2. JavaScript may check that the input is not empty and then send the information to the server.
3. The server receives the request.
4. The server checks the account information, often using a database.
5. The server sends a response.
6. The browser displays the result, such as a dashboard or an error message.

Basic security idea

Real websites must protect user data. Common security topics include HTTPS, strong authentication, access control, input validation, secure password storage, software updates, and protection against common web attacks. Security is an important part of professional web development.

12. Beginner Learning Roadmap

Stage	Topic	What to Learn
1	HTML Basics	Learn headings, paragraphs, links, images, lists, tables, forms, and semantic elements.
2	CSS Basics	Learn colors, fonts, box model, borders, positioning, Flexbox, Grid, and responsive design.
3	JavaScript Basics	Learn variables, data types, operators, conditions, loops, functions, arrays, events, and DOM manipulation.
4	Small Projects	Create a profile page, calculator, registration form, to-do list, or simple portfolio.
5	Git	Use version control to save and manage changes to your projects.
6	Back-End	After front-end basics, learn a server-side language/framework and databases.
7	Full Projects	Connect front end, back end, APIs, and database into complete applications.

13. Quick Revision

- **Web development:** creating and maintaining websites and web applications.
- **HTML:** structure and content.
- **CSS:** appearance and layout.
- **JavaScript:** behavior and interaction.
- **Browser:** software that requests and displays web resources.
- **Server:** system that responds to requests and provides resources/data.
- **DNS:** helps translate domain names into IP addresses.
- **HTTP/HTTPS:** protocols used for Web communication.
- **Tim Berners-Lee:** inventor of the World Wide Web.
- **Internet vs Web:** the Internet is the underlying network; the Web is a system of linked resources and applications that uses the Internet.

Final idea

If you remember only one concept, remember this: **HTML builds the structure, CSS makes it look good, and JavaScript makes it interactive.** These technologies work inside a larger system involving browsers, servers, networks, DNS, databases, APIs, and security.